



Azure Master Architect Capstone Project + Microsoft Fabric Rayfin

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Azure Master Architect (AMA) Program – Overview

An invite-only, cohort-based development program for Microsoft Azure architects, advancing architecture mastery, customer impact and technical leadership at enterprise scale.



THE FOCUS

What it strengthens

- Deep technical architecture skills
- Measurable customer impact
- Technical leadership at enterprise scale



THE JOURNEY

How it works

- A structured, cohort-based path
- Advanced architecture training + mentorship
- A hands-on capstone solution
- Culminates in a customer-ready executive presentation



THE OUTCOME

What graduates can do

- Design, defend & deliver complex Azure solutions
- Influence technical strategy
- Lead architectural excellence across customers, regions & communities



Business Case

Autonomous Multi-Modal Logistics Agent Network

AI agents dynamically optimizing freight routes across transportation modes and autonomously responding to disruptions Advantage: 20% cost reduction; 30% faster delivery; carbon footprint reduction

- **Route Planning Agent:** Calculating optimal paths considering cost, time, capacity across modes
- **Real-Time Optimization Agent:** Dynamically rerouting shipments based on traffic, weather, delays
- **Carrier Selection Agent:** Autonomously booking with optimal carriers via API integrations
- **Carbon Tracking Agent:** Measuring and minimizing environmental impact of routes
- **Customer Communication Agent:** Proactively updating customers on delivery status

Success Criteria

Reduce freight costs by 20% across all shipments through AI optimization

Improve on-time delivery rate from 85% to 98% via predictive agent planning

Reduce carbon emissions per shipment by 30% through mode optimization

Optimize carrier utilization to 90%+ capacity via intelligent load matching

Provide customers accurate delivery windows within 1 hour through agent tracking

Regional Specificity

How will the multi-modal logistics agent network adapt to Asia's diverse customs regulations, varying infrastructure capabilities across countries (e.g., India's road networks, China's high-speed rail), multilingual carrier communications, and cross-border data sharing restrictions while dynamically optimizing routes and reducing carbon emissions? Assume scenario details and document architectural decisions.

ASIA AMA Program Graduates

ANZ



David Watson



Francesco Fava



John Billiris



Katherine Tran



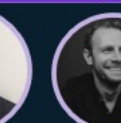
Minette Steynberg



Vic Perdana



Adam Strickland



Arian Nevjestic



Clarence Bakirtzidis



David Jansen



Kotomi Uchiyama



Richard Mackrill



Richard Smith



Sergio Zenatti Filho

ASEAN



Brendon Matheson



Alain Vetier



Francis Joseph



Gopi Sundaresan



Weixian Zhang

INDIA & SOUTH ASIA



Ajay Barun



Chetan Vaja



Deep Amberkar



Sanjeev Kumar



Saurabh Seth



Shishir Garde



Vidya Narasimhan

GCR



Max Zhou



Daniel Wang



Frankie Lai



Xuebing Bai

KOREA



Haewon Shin



Inhwan Hwang



TagHwan Kim



Yong Hee Park

ASIA DN HQ



Yun Kyoung Aum

JAPAN



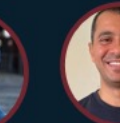
Junya Fujimoto



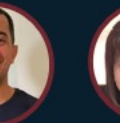
Toru Makabe



Yuta Tatewaki



Mohamed Mostafa



Kanako Tokushige



Issei Hiraoka



Hideyuki Nishiyori

Top 5 Architects



Weixian Zhang



Sergio Zenatti Filho



Richard Mackrill



Kotomi Uchiyama

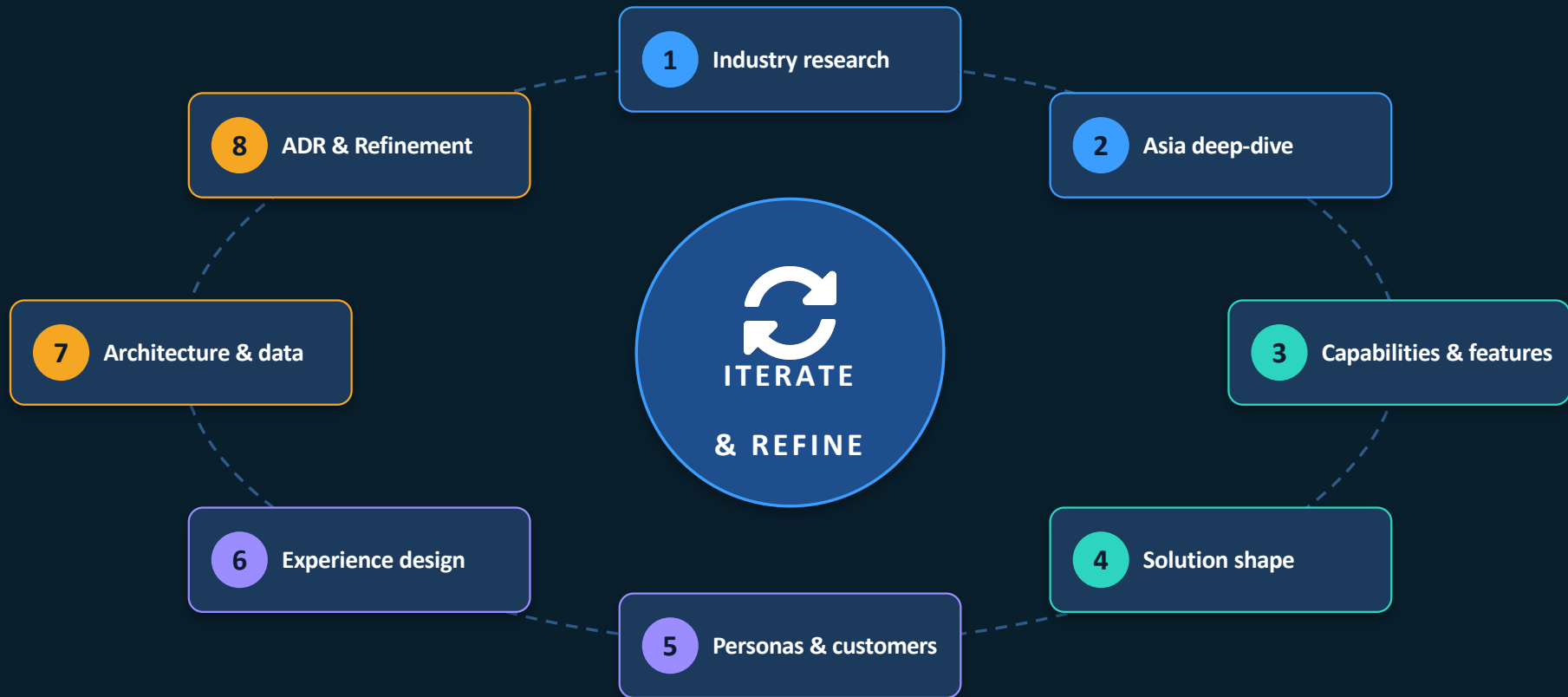


Max Zhou





A Cyclic Design Method, Refined Over Time



Clockwise through four phases — and continuously looping: what we learn building feeds straight back into discovery.

● Discover ● Define ● Design ● Architect & Refine



Tooling & Spec-Driven Development

1

STEP 1

AI Pair-Programming in VS Code

Built the solution in VS Code with GitHub Copilot as a constant pairing partner, using AI assistance for scaffolding, refactors and debugging across the entire stack.

IN PRACTICE

- **IDE:** VS Code + GitHub Copilot, used end-to-end.
- **Coverage:** Backend, frontend, infrastructure-as-code & tests.
- **Habit:** Inline suggestions plus chat-driven refactors.

OUTPUT *A faster, AI-assisted iteration loop across the codebase.*

2

STEP 2

Spec-Driven Development with Spec Kit

Adopted Spec Kit to turn requirements into structured, versioned specs before code, keeping implementation traceable back to intent.

IN PRACTICE

- **Workflow:** A 12-step spec-driven development cycle.
- **Records:** Every decision captured as an ADR or RFC.
- **Benefits:**
 - Specs stayed the source of truth
 - Consistent structure per feature
 - AI agents grounded in the spec
 - Clear audit trail, idea to code

OUTPUT *A structured, auditable path from idea to implementation.*



Agent “Battle”: Cross-Model Validation

3

STEP 3

GPT Agent — Analysis & Proposals

Ran GPT-based models as a first line of AI agents, analysing issues in the build, drafting fixes and proposing approaches for the harder design decisions.

IN PRACTICE

- **Role:** First-pass issue analysis & solution drafts.
- **Habit:** Proposed fixes and alternative approaches.
- **Style:** Fast, broad, sometimes over-confident.

OUTPUT *An independent first opinion on every issue.*

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STEP 4

Claude Agent — Critique & Validation

Ran Claude-based models in parallel as a second, independent reviewer, cross-checking GPT’s reasoning and validating the approach before it shipped.

IN PRACTICE

- **Role:** Independent critique & validation of proposals.
- **Habit:** Challenged assumptions, flagged edge cases.
- **Validation checklist:**
 - Logic consistency
 - Edge-case coverage
 - Security & compliance gaps
 - Final sign-off before merge

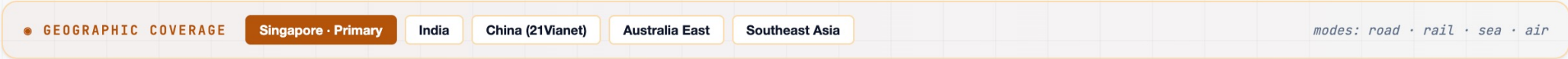
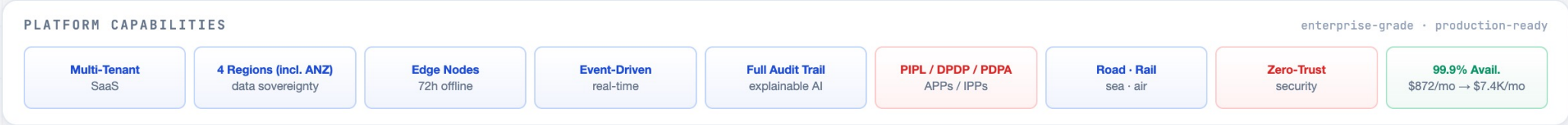
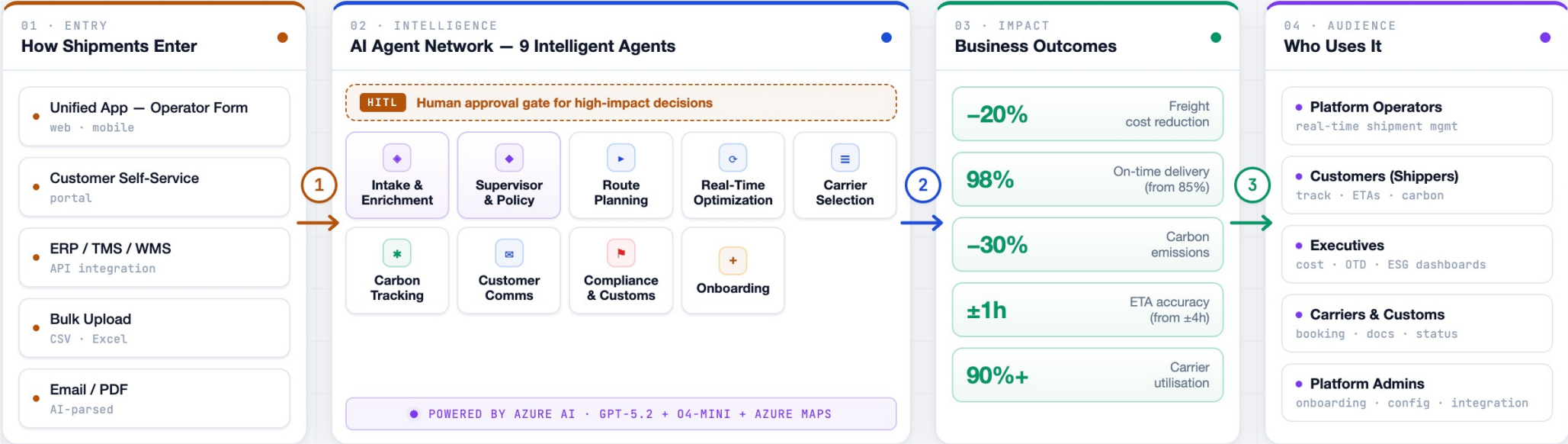
OUTPUT *Higher-confidence decisions, fewer blind spots.*

Autonomous Multi-Modal Logistics Platform

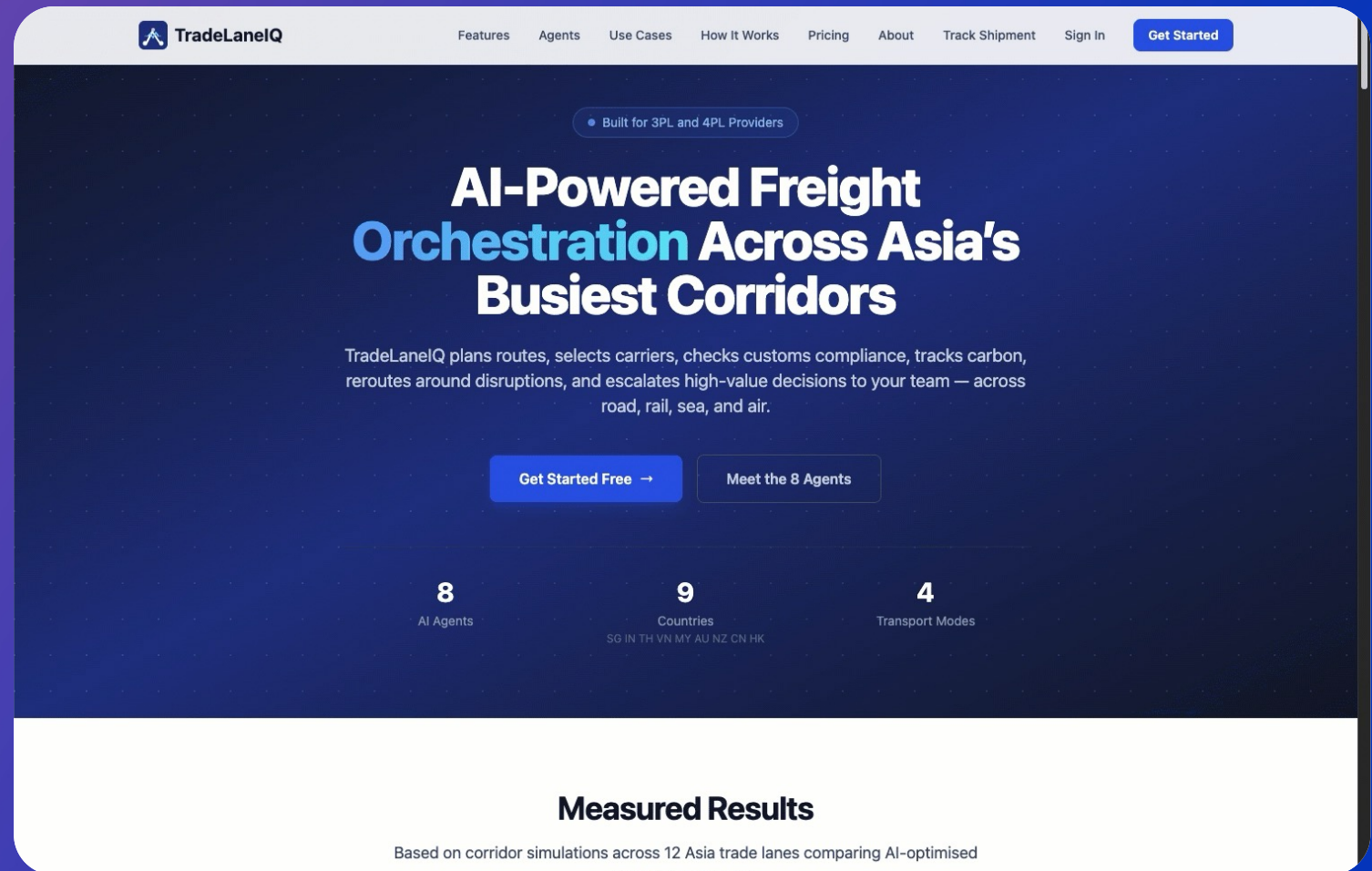
AI-native, multi-tenant SaaS control tower for cross-border freight orchestration in Asia-Pacific

MICROSOFT AZURE

9 AI AGENTS · 4 REGIONS



Demo



TradeLanelQ Control Tower

+

https://dev.tradelanelq.com

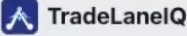
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TradeLanelQ

FeaturesAgentsUse CasesHow It WorksPricingFAQAboutTrack ShipmentDashboard

Built for 3PL and 4PL Providers

AI-Powered Freight Orchestration Across Asia's Busiest Corridors

TradeLanelQ plans routes, selects carriers, checks customs compliance, tracks carbon, reroutes around disruptions, and escalates high-value decisions to your team — across road, rail, sea, and air.

Get Started Free →

Book a Demo

Meet the 9 Agents

9
AI Agents

9
Countries
SG IN TH VN MY AU NZ CN
HK

12
Corridors

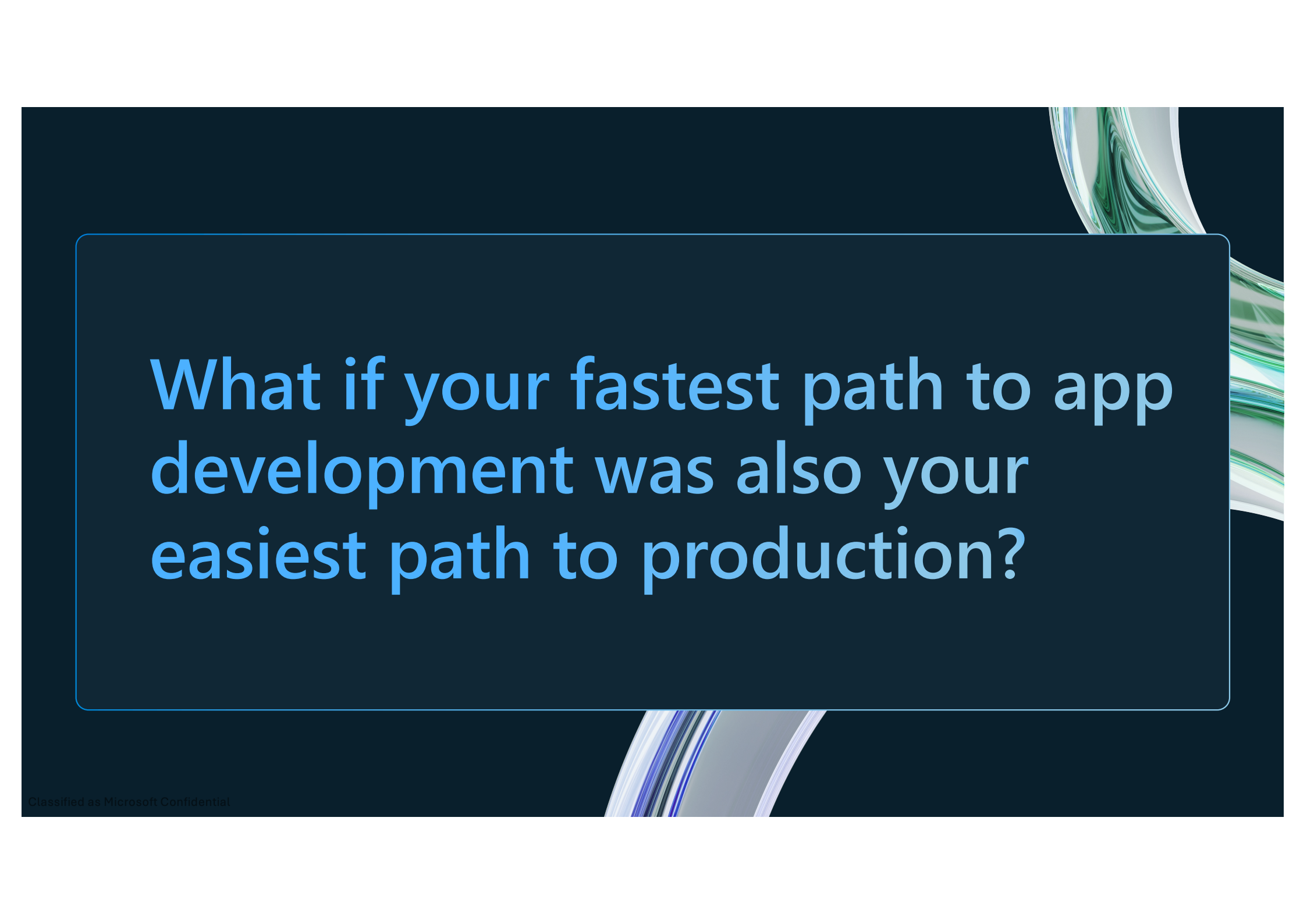
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Transport Modes

Measured Results

Based on corridor simulations across 12 Asia trade lanes comparing AI-optimised

Requiem

A code-first backend for the agentic era.



What if your fastest path to app development was also your easiest path to production?

**Every developer
workflow is becoming
agent-powered**

76%

Increase in agentic coding in 2025¹



**But enterprise requirements are
keeping most agent-authored
apps from production**

80-90%

of AI-generated apps never reach production
due to enterprise readiness gaps²

1. Stack Overflow's 2025 Developer Survey

2. IDC, Everest Group, developer studies

Real apps in the enterprise need more than vibes



Software developers need to:

- Stand up a databases
- Wire up authentication
- Configure backend services
- Safely connect to existing customer data
- Follow compliance and governance rules



IT admins need to:

- Keep one view of company data
- Give the right people the right data access
- Maintain security and compliance
- Keep cloud costs in check
- Prove it all to auditors

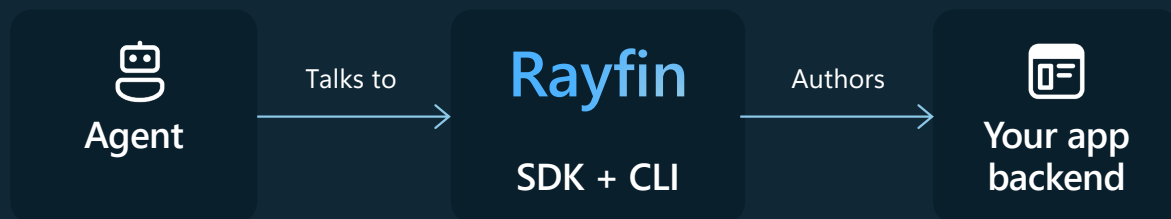
Public preview

Rayfin

Build apps on
Microsoft Fabric
with a managed,
code-first backend

Vibe.

Build it with agent-friendly Rayfin



Deploy.

Deploy it on enterprise-ready Fabric



Repeat.

Accelerate time to market with a reimagined app dev cycle

— EXTEND

Templates.

Don't start from scratch. A template bundles everything an agent needs to build a specific kind of app — instructions, tools, code, config, libraries.

What every template ships

- 01 **Agent Instructions**
Project-level instructions and skills the agent reads before touching the codebase.
- 02 **MCP servers / API definitions**
Domain-specific tools the agent can call — schema lookups, dataset browsers, generators.
- 03 **Sample code**
A working starter screen in the language and idioms of the target app type.
- 04 **Rayfin configuration (rayfin.yml)**
Auth providers, deployment target, DB dialect — pre-configured.
- 05 **Libraries**
Pre-wired SDKs, components, helpers — typed and ready on first run.

Data App

Build apps on top of your semantic models.

- Data package — instructions on working with semantic models
- Charting libraries pre-wired with the Rayfin client
- Visuals that look good on day one
- Best for · dashboards, insights, BI tooling on top of your data

```
$ npm create @microsoft/rayfin@latest -- "Pulse" --template data-app --workspace "Sales Insights"
```

— TRY IT

Build your first Rayfin app today.

```
terminal • LIVE  
$ npm create @microsoft/rayfin@latest "pulse"  
$ cd pulse  
$ npx rayfin up
```

[Get Started](#)



<https://github.com/szenatti/presentations>

Thank you